DATA ENGINEERING PREPARATION – DAILY PRACTICE

<u>DAY 3 PRACTICE – PYTHON</u>

Exercise 1 – Sum of Even Numbers

```python
---
'''
# Exercise 1 – Sum of Even Numbers

numbers = [3, 8, 12, 5, 7, 10]
sum = 0
for i in numbers:
    if i % 2 == 0:
        sum += i
print(f"Sum of Even Number : {sum}")
```

3] ✓ 0.0s

```
Sum of Even Number : 30
```

Exercise 2 – Count Vowels in a String

```python
# Exercise 2 - Count Vowels in a String

word = input("Enter your word:")
list = ['a', 'e', 'i', 'o', 'u']
count = 0
for i in word:
    if i in list:
        count += 1
print(f"Input : \'{word}\' \nNumber of vowels : {count}")
```

10] ✓ 3.1s

```
Input : 'education'
Number of vowels : 5
```

Exercise 3 – Find Minimum Number

```
---
'''
# Exercise 3 - Find Minimum Number
numbers = [45, 12, 78, 3, 25]

min_value = numbers[0]
print(min_value)

for i in numbers[1:]:
    if i < min_value:
        min_value = i

print(f"Minimum Number: {min_value}")
```

[15] ✓ 0.0s

```
45
Minimum Number: 3
```

Exercise 4 – Square of Numbers

```
--
'''
# Exercise 4 - Square of Numbers
lst = [1,2,3,4,5]
square = [x**2 for x in lst]
print(f"Square of Numbers List : {square}")
```

[18] ✓ 0.0s

```
Square of Numbers List : [1, 4, 9, 16, 25]
```

Exercise 5 – Character Frequency

```
'''
# Exercise 5 – Character Frequency
word = input("Enter a word: ")
frqe = {}
for i in word:
    if i in frqe:
        frqe[i] += 1
    else:
        frqe[i] = 1
for key, value in frqe.items():
    print(key ,':', value)
```

[48] ✓ 2.6s

```
...
a : 1
p : 2
l : 1
e : 1
```

Exercise 6 – Remove Negative Numbers

```
'''
# Exercise 6 – Remove Negative Numbers
numbers = [5, -3, 8, -1, 9, -6]

new = [x for x in numbers if x > 0]
print(f"positive numbers list is : {new}")
```

[29] ✓ 0.0s

```
... positive numbers list is : [5, 8, 9]
```

Exercise 7 – Student Marks Dictionary

```
'''
# Exercise 7 - Student Marks Dictionary
marks = {
    "Rahul": 85,
    "Amit": 72,
    "Priya": 90,
    "Neha": 66
}
for key,value in marks.items():
    if value > 80:
        print(key)
```

[33] ✓ 0.0s

```
Rahul
Priya
```

Exercise 8 – List Length Without len()

```
'''
# Exercise 8 - List Length Without len()
numbers = [10,20,30,40,50]

count = 0
for i in numbers:
    count += 1
print(f"Length of list : {count}")
```

[43] ✓ 0.0s

```
Length of list : 5
```

Exercise 9 – Check Palindrome

```
'''
# Exercise 9 - Check Palindrome
word = input("Enter a word: ")

if word == word[::-1]:
    print(f" \'{word}\' word is a palindrome.")
else:
    print(f" \'{word}\' word is not a palindrome.")
```

[41] ✓ 2.7s

```
… 'madam' word is a palindrome.
```

Exercise 10 – Multiples of 3 and 5

```
---

'''
# Exercise 10 - Multiples of 3 and 5
for i in range(1, 50):
    if i % 3 == 0 and i % 5 == 0:
        print(i)
```

[42] ✓ 0.0s

```
… 15
  30
  45
```

LINUX PRACTICE

Perform the following tasks in terminal:

1. Create a directory called data_files.

2. Inside the directory create three files:

   sales.txt, users.txt, logs.txt

```
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise$ ls -lrth
total 4.0K
drwxrwxrwx 1 prashant prashant 4.0K Mar  6 20:20 practice_day1
-rwxrwxrwx 1 prashant prashant  876 Mar  6 20:25 numbers.txt
drwxrwxrwx 1 prashant prashant 4.0K Mar  6 20:42 logs
drwxrwxrwx 1 prashant prashant 4.0K Mar  7 17:28 data_files
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise$ cd data_files/
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_files$ touch sales.txt users.txt logs.txt
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_files$ ls -lrth
total 0
-rwxrwxrwx 1 prashant prashant 0 Mar  7  2026 users.txt
-rwxrwxrwx 1 prashant prashant 0 Mar  7  2026 sales.txt
-rwxrwxrwx 1 prashant prashant 0 Mar  7  2026 logs.txt
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_files$
```

3. Display contents of sales.txt.

4. Display first 3 lines of users.txt.

5. Display last 2 lines of logs.txt.

```
2026-03-06 09:28:05 INFO        AUTH            User logout
EOF
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_files$ cat sales.txt
SALE_ID DATE       PRODUCT  CUSTOMER_ID QUANTITY PRICE STATUS
S101    2026-03-01 Laptop   U201        1        55000 SUCCESS
S102    2026-03-01 Mouse    U202        2        500   SUCCESS
S103    2026-03-02 Keyboard U203        1        1200  SUCCESS
S104    2026-03-02 Monitor  U204        1        15000 FAILED
S105    2026-03-03 Laptop   U205        1        56000 SUCCESS
S106    2026-03-03 Printer  U201        1        8000  SUCCESS
S107    2026-03-04 Mouse    U206        3        500   SUCCESS
S108    2026-03-04 Keyboard U207        1        1200  FAILED
S109    2026-03-05 Laptop   U202        1        54000 SUCCESS
S110    2026-03-05 Monitor  U208        1        15500 SUCCESS
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_files$ head -n 3 users.txt
USER_ID NAME   CITY   AGE STATUS
U201    Rahul  Mumbai 28  ACTIVE
U202    Sneha  Pune   25  ACTIVE
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_files$ tail -n 2 logs.txt
2026-03-06 09:26:30 ERROR       SERVER          Service crashed
2026-03-06 09:28:05 INFO        AUTH            User logout
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_files$
```

SQL PRACTICE

Table: employees

Columns: emp_id, name, department, salary

[SQL Worksheet]*

```sql
CREATE TABLE EMPLOYEES2(emp_id NUMBER, name VARCHAR2(20), department VARCHAR2(20), salary NUMBER);
INSERT INTO EMPLOYEES2 VALUES (101, 'Rahul', 'HR', 35000);
INSERT INTO EMPLOYEES2 VALUES (102, 'Sneha', 'Finance', 42000);
INSERT INTO EMPLOYEES2 VALUES (103, 'Amit', 'IT', 60000);
INSERT INTO EMPLOYEES2 VALUES (104, 'Priya', 'Sales', 38000);
INSERT INTO EMPLOYEES2 VALUES (105, 'Rohit', 'Marketing', 41000);
INSERT INTO EMPLOYEES2 VALUES (106, 'Anjali', 'IT', 65000);
INSERT INTO EMPLOYEES2 VALUES (107, 'Karan', 'HR', 36000);
INSERT INTO EMPLOYEES2 VALUES (108, 'Meena', 'Finance', 43000);
INSERT INTO EMPLOYEES2 VALUES (109, 'Vijay', 'IT', 70000);
INSERT INTO EMPLOYEES2 VALUES (110, 'Pooja', 'Sales', 39000);
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.008 seconds

	EMP_ID	NAME	DEPARTMENT	SALARY
1	101	Rahul	HR	35000
2	102	Sneha	Finance	42000
3	103	Amit	IT	60000
4	104	Priya	Sales	38000

Write SQL queries to:

1. Display all employees working in department 'IT'.

[SQL Worksheet]*

```sql
-- 1. Display all employees working in department 'IT'.
SELECT * FROM EMPLOYEES2 WHERE DEPARTMENT = 'IT';
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.005 seconds

	EMP_ID	NAME	DEPARTMENT	SALARY
1	103	Amit	IT	60000
2	106	Anjali	IT	65000
3	109	Vijay	IT	70000

2. Display employees whose salary is greater than 60000.

```sql
-- 2. Display employees whose salary is greater than 60000.
SELECT * FROM EMPLOYEES2 WHERE SALARY > 60000;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download ▾ Execution time: 0.001 seconds

	EMP_ID	NAME	DEPARTMENT	SALARY
1	106	Anjali	IT	65000
2	109	Vijay	IT	70000

3. Display employees sorted by salary in ascending order.

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[SQL Worksheet]* ▾   Aa ▾

```sql
-- 3. Display employees sorted by salary in ascending order.
SELECT * FROM EMPLOYEES2 ORDER BY SALARY ASC;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download ▾ Execution time: 0.004 seconds

	EMP_ID	NAME	DEPARTMENT	SALARY
1	101	Rahul	HR	35000
2	107	Karan	HR	36000
3	112	Neha	HR	37000
4	104	Priya	Sales	38000
5	110	Pooja	Sales	39000
6	114	Kavita	Sales	40000

4. Find the highest salary in the table.

[SQL Worksheet]*

```sql
-- 4. Find the highest salary in the table.
SELECT * FROM EMPLOYEES2 ORDER BY SALARY DESC FETCH FIRST 1 ROW ONLY;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download   Execution time: 0.006 seconds

	EMP_ID	NAME	DEPARTMENT	SALARY
1	109	Vijay	IT	70000

5. Count total number of employees.

[SQL Worksheet]*

```sql
-- 5. Count total number of employees.
SELECT COUNT(*) FROM EMPLOYEES2;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download   Execution time: 0.002 seconds

	COUNT(*)
1	15